



# **Deep learning**

## **CSE5DL**

**LMS Subject link:**

**<https://lms.latrobe.edu.au/course/view.php?id=147232>**

**Student cohort**

## **SUBJECT LEARNING GUIDE**

**Teaching Period Semester 1, 2024**

**Campus**

**Bundoora**

**Subject Coordinator:           Zhen He**

**Instance Coordinator:       Zhen He**

# SUBJECT DETAILS

**Subject Code:** CSE5DL  
**Subject Title:** Deep Learning  
**Teaching Period:** Semester 1  
**Location(s):** TLC-02-214  
**Credit Points:** 15  
**Subject Delivery Mode:** Face to Face  
**AQF Level:** Postgraduate  
**Subject relationships:**

## ENROLMENT REQUIREMENTS

**Prerequisites:** CSE100F OR CSE5APG OR CSE4IP OR CSE1CPP OR CSE400F  
**Co-requisites:**  
**Assumed Skills & Knowledge:** Programming skills in some programming language  
**Special Study Requirements:**

**Required resources:**

**Additional subject requirements:**

**Class materials:**

## STAFF CONTACTS

**Subject Coordinator:** Associate Professor Zhen He  
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**Additional contact details:**  
**Location:** BG 235

**Tutor:** Josh Millward  
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**Tutor:** Albert Bonela  
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## ABOUT THIS SUBJECT

Deep learning is currently the central machine learning method fuelling the artificial intelligence revolution. In this subject you learn how to apply deep learning algorithms to solve real-world problems. This subject does not assume you have previous machine learning experience, therefore it starts teaching deep learning at a very introductory level. You learn how deep learning techniques can be applied to such tasks as image recognition, sentiment classification, machine translation, question and answering, speech synthesis, etc. The practical skills taught in this subject will allow you to build production level deep learning software that can scale out to millions of users. You will be introduced to the popular deep learning programming frameworks of Pytorch and Tensorflow and advanced deep learning techniques such as reinforcement learning, generative adversarial networks and few shot learning.

## SUBJECT INTENDED LEARNING OUTCOMES (SILOs)

Upon successful completion of this subject, you should be able to:

1. Analyse the advantages and disadvantages of using traditional machine learning algorithms versus deep learning algorithms for solving industry problems.
2. Analyse a given industry problem and then recommend which deep learning algorithm best solves the problem. This requires the student to know which best learning algorithm is most suitable to solve each type of problem.
3. Write/produce Pytorch code to implement deep learning algorithms to solve computer vision and natural language processing problems.
4. Propose deployment and maintenance strategies for deep learning production systems in the cloud based on the analysis from the machine learning requirements of a business.
5. Given a real-world problem where labels are scarce, investigate different deep learning approaches to tackle this problem.
6. Design the architecture of a generative adversarial network that can generate realistic data.

## GRADUATE CAPABILITIES (GC)

| GC                              | GC Elements |
|---------------------------------|-------------|
| Communication                   | capstone    |
| Inquiry and Analysis            | capstone    |
| Personal and Professional       | capstone    |
| Discipline Knowledge and Skills | CSE5DL      |

## LEARNING ACTIVITY SCHEDULE

| Week/<br>Session | Date | Learning Activity                        | Contact<br>hours |
|------------------|------|------------------------------------------|------------------|
| 1                | 5/3  | Lecture 1: Introduction to Deep Learning | 2                |
| 2                | 12/3 | Lecture 2: Foundations of Deep Learning  | 2                |

|              |      |                                                                                   |           |
|--------------|------|-----------------------------------------------------------------------------------|-----------|
| 3            | 19/3 | Lecture 3: Deep Learning for computer vision                                      | 2         |
| 4            | 26/3 | Lecture 4: Practical Deep Learning                                                | 2         |
| 5            | 9/4  | Lecture 5: Deep Learning for Natural Language Processing                          | 2         |
| 6            | 16/4 | Lecture 6: Natural Language Processing in Practice                                | 2         |
| 7            | 23/4 | Lecture 7: Deep Learning in Production                                            | 2         |
| 8            | 30/4 | Lecture 8: Unsupervised Deep Learning                                             | 2         |
| 9            | 7/5  | Lecture 9: Deep learning for time series analysis and deep reinforcement learning | 2         |
| 10           | 14/5 | Lecture 10: Advanced Computer Vision                                              | 2         |
| 11           | 21/5 | Lecture 11: Graph neural networks, domain adaptation                              | 2         |
| 12           | 28/5 | Lecture 12: Exam revision                                                         | 2         |
| <b>Total</b> |      |                                                                                   | <b>24</b> |

\*Timetable details are available at <http://www.latrobe.edu.au/students/timetables>. Students should also regularly check this subject's LMS site for updated information.

#### Details of any self-allocation activities:

## SELF-DIRECTED STUDY

In addition to the learning activities specified, you are expected to allocate approximately **4 hours per week of self-directed study in this subject**. This time includes subject preparation, completing learning activities, readings, assessment preparation and other related work to enhance your participation in this subject and your learning at La Trobe.

## LEARNING RESOURCES & TECHNOLOGIES

| Resource Type    | Title                      | Author and Year                                         | Publisher            |
|------------------|----------------------------|---------------------------------------------------------|----------------------|
| Recommended book | Deep Learning              | Ian Goodfellow, Yoshua Bengio and Aaron Courville, 2016 | MIT Press            |
| Recommended book | Deep learning with PyTorch | Eli Stevens, Luca Antiga and Thomas Viehmann, 2020      | Manning Publications |

## ASSESSMENT AND FEEDBACK SUMMARY

| Week/Session | Date due*        | %  | Assessment Activity    | Feedback method | ILOs assessed |
|--------------|------------------|----|------------------------|-----------------|---------------|
| 1.           | Sunday following | 10 | Quizzes linked to labs |                 | 3,5           |

|    |                        |    |            |               |             |
|----|------------------------|----|------------|---------------|-------------|
|    | the week of<br>the lab |    |            |               |             |
| 2. | 31/5/2024              | 40 | assignment | Marking sheet | 3           |
| 3. | Exam<br>Period         | 50 | exam       | Exam mark     | 1,2,3,4,5,6 |

#### Additional information about assessment & feedback

Students should keep a copy of all assignments submitted for assessment.

## MANDATORY REQUIREMENTS

There are a variety of assessment hurdles or conditions for passing a subject other than the overall mark. **Mandatory Requirements** may be built into the assessment of the learning outcomes of your subject.

**There are no Hurdle Requirements for this subject / The Hurdle Requirements for this subject are:**

**Other mandatory requirements for this subject are:**

## GENERAL ASSESSMENT REQUIREMENTS

### SUBMITTING WRITTEN WORK WITH TURNITIN

**Turnitin** is a web-based text-matching software system used at La Trobe University to assist you in writing your assignments and checking for similarity with existing published work. Please remember to allow adequate time to submit your assignment to Turnitin. **It is your responsibility to have your assignment submitted by the assessment due date.** Not receiving an originality report is not an acceptable reason for requesting an assignment extension unless it is a required part of the assessment.

### POLICIES, PROCEDURES AND GUIDELINES

The University has a comprehensive policy framework to which both staff and students must adhere. You should familiarise yourself with those policies, procedures and guidelines likely to affect you, especially the following:

- [Academic Progress Review Policy](#)
- Assessment Policy
- [Health and Safety](#) (OHS)
- [Privacy](#)
- [Student Academic Misconduct Policy](#)

## ADJUSTMENTS TO ASSESSMENT

You may be affected by a range of adverse circumstances while you are preparing for or undertaking an assessment task. There are avenues for adjustments to your assessment depending on the circumstances you face including opportunities to apply for extensions to submission dates, Special Consideration and Alternative Assessment Schedules. For more detailed information refer to the [Assessment Procedure – Adjustments](#).

## REQUEST AN EXTENSION OF TIME TO SUBMIT AN ASSIGNMENT TASK

Where you know in advance that you will not be able to submit an assessment task by the due date due to adverse circumstances that have affected you during the preparation of the task, you will need to [request an extension of time to submit](#). **This must be done at least three days prior to the due date.** Penalties normally apply if you submit an assessment after the due date. **Poor time management is not an acceptable reason for an extension.**

To apply, go to <https://www.latrobe.edu.au/students/admin/forms/request-an-extension/request>

## SPECIAL CONSIDERATION

If you have experienced serious short term, adverse and unforeseen circumstances that substantially affect your ability to complete an assessment task to the best of your potential, you may be eligible to apply for **Special Consideration**. To do this or find more information, go to <http://www.latrobe.edu.au/special-consideration>

## STUDENTS WITH A LEARNING ACCESS PLAN (LAP)

Students who have a Learning Access Plan must apply for any adjustments they may need via the same channels but may have access to more flexible arrangements depending on their circumstances. Please refer to the [Assessment Procedure – Adjustments](#) and your LAP.

## REVIEW AND RE-MARK

For information on what to do if you wish to seek a review of an assessment outcome see Part B of the Validation and Moderation Procedure.

## ACADEMIC INTEGRITY

Academic integrity means being honest in academic work and taking responsibility for learning the conventions of scholarship. Academic integrity education is integral to the learning experience at La Trobe University. The University requires its academic staff and students to observe the highest ethical standards in all aspects of academic work, and it demonstrates its commitment to these values by awarding due credit for honestly conducted scholarly work, and by penalising academic misconduct and all forms of cheating.

The penalty for submitting an assignment under your name that is the work of a third party may be severe, even leading to exclusion from the University without readmission. Refer to the **Academic Integrity - Schedule of Penalties and Actions** within the [Student Academic Misconduct Policy](#).

You should understand and be able to practice integrity in all your academic work, be familiar with the [Academic Integrity website](#) and complete the **Academic Integrity Module (AIM)** found in your LMS.

If you have any specific questions about **academic integrity** and your assessment, your Subject or Course Coordinator will be able to assist.

## STUDENT FEEDBACK ON SUBJECT SURVEYS

The **Student Feedback on Subjects (SFS) Survey** is one of the quality assurance processes that occur across the University. In this survey, you can tell us about your learning experiences in this subject. We value your feedback. Your views are taken seriously and assist us to improve the learning experience for future students through the LTU Subject Monitoring process.

The surveys will be available for you to complete towards the end of the subject teaching period. Feedback from previous students who completed this subject is available in the LMS site for this subject.

For more information you may contact [surveys@latrobe.edu.au](mailto:surveys@latrobe.edu.au).

## CHARTER OF STUDENT RIGHTS AND RESPONSIBILITIES

As a La Trobe coursework student, you have rights and responsibilities which are governed by La Trobe statutes and regulations, as well as policies and procedures. La Trobe University aspires to provide an outstanding student experience which is conducive to scholarly activity and supportive of your learning goals.

The **Charter of Student Rights and Responsibilities** is a summary of your rights and responsibilities as a student at La Trobe. The Charter was developed in consultation with La Trobe students. <https://policies.latrobe.edu.au/document/view.php?id=225>.

## STUDENT COMPLAINTS

The **Student Complaints Management System (SCM)** is a formal mechanism for students to lodge concerns about any aspect of their university experience. A student complaint may be about an administrative matter, a problem with a subject or course, or the conduct of a student or staff member.

For more information about the **Student Complaint process** and how to lodge a complaint, visit the **Complaints and Feedback** page at: <https://www.latrobe.edu.au/students/support/complaints>.

# GETTING HELP AND SUPPORT AT UNI

A range of services at La Trobe can assist you with your studies, health and wellbeing.

For the full range of services and facilities available for you, go to **Student Help and Support** at <http://www.latrobe.edu.au/students/support>.

## STUDYING ONLINE

When you are in studying an online subject, please be sure that you can continue to access our learning support services through the Learning Hub (academic skills, English language, Maths and Science) via Zoom appointments that you can book by accessing the [Learning Hub LMS page](#) or by emailing the [LearningHub@latrobe.edu.au](mailto:LearningHub@latrobe.edu.au).

You also have access to assignment feedback and online study support chat through Studiosity that can be accessed from any subject [LMS](#) page. You can also access the Library services remotely through [the library website](#).

## LIBRARY

Your **Library** offers many ways to support your study. To discover more about how the Library can help, visit your **Library website** and explore the following essential resources:

- **Achieve@Uni**  
<https://www.latrobe.edu.au/students/study-resources/learning/achieve> - Learn skills to help complete your assessments
- **Assessment Planner**  
<http://www.latrobe.edu.au/library/assessment-thesis-support/plan-my-assessment> - Create a personal assessment schedule
- **Expert Help Guides**  
<https://latrobe.libguides.com/> - Find expert discipline and specialist help for your research
- **Referencing at La Trobe University**  
<http://latrobe.libguides.com/referencing/> - Understand how and why to reference
- **Academic Referencing Tool**  
[www.lib.latrobe.edu.au/referencing-tool](http://www.lib.latrobe.edu.au/referencing-tool) - Get detailed examples of referencing styles
- **Chat to library staff**  
[www.latrobe.edu.au/library](http://www.latrobe.edu.au/library) - Access the text chat service from the Library website
- **Contact us**  
[www.latrobe.edu.au/library/about/contact](http://www.latrobe.edu.au/library/about/contact) - Call or email your Library

## ASK LA TROBE

**ASK La Trobe** is a service open to all La Trobe students who need guidance with administrative matters during their time at La Trobe. ASK La Trobe can provide you with help relating to enrolments and



timetabling, fees and financial support, exams, results and graduations & student life and support services.

Go to **ASK La Trobe – Current Students** at <http://latrobe-current.custhelp.com/app/home> and start by looking at our **FAQs** or visit us at your campus **ASK La Trobe Help Zone**.

**If you can't find what you're looking for:**

- **Call** 1300 La Trobe (1300 528 762)
- **Ask a question** online at <http://latrobe-current.custhelp.com/app/ask>
- **Visit** the **ASK La Trobe Help Zone**. Join the queue for the Bundoora, Bendigo or city campuses using the **QLess: Virtual Queuing system**.

## LEARNING AND SUPPORT PROGRAMS

There are many resources, including services, to help you with your academic studies at La Trobe. Go to the **Learning and Support Programs** page at <http://www.latrobe.edu.au/students/study-resources/learning> for further information.

## HEALTH AND WELLBEING

Your health and wellbeing are an important part of being a successful student. Studying can be stressful and many factors can affect your mental and physical health, in and outside of the University.

Check out the services available to you at La Trobe to support your health and wellbeing at [www.latrobe.edu.au/wellbeing](http://www.latrobe.edu.au/wellbeing).